

Shubham Kale

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EDUCATION

Johns Hopkins University

Expected May 2028

B.S. Computer Science, B.S. Applied Mathematics & Statistics, B.A. Economics | GPA: 3.70/4.00 *Baltimore, MD*

- **Coursework:** Data Structures & Algorithms, OOP (Java), Linear Algebra for Data Science, Statistical Analysis.

Honors: Hodson Trust Scholar (from 45,000+ applicants), 1For2 Foundation Scholar (5 of 1,000+)

EXPERIENCE

Summer Atlantic Capital LLC & SAC Nexus

Jun 2026 – Present

Software Engineer, Product Development Team

Remote

- Built and shipped a compliance-scoring layer across **4 financial-risk workflows** (KYC, transactions, escrow, AML), cutting KYC case-reconstruction time by **1–3 minutes per review** and transaction review by **1–2 minutes per payment** — roughly **4 analyst-hours/day** across 80 daily scored cases
- Developed an **AML candidate-ranking and explanation system** that compressed a 45-minute partner-screening workflow to **15–20 minutes (a 56–67% reduction)**, saving **4–5 analyst-hours per week per analyst**; modeled at **~\$20K/year** of labor capacity per analyst, or **~\$200K/year** across a 10-analyst deployment
- Implemented **fail-closed escrow routing** that eliminated a false-safe failure mode where missing risk inputs scored 0 and could yield a misleading Low-risk result
- Built the service from an empty repository: **7 REST endpoints** on a versioned **FastAPI microservice** over a modular pipeline (request → entity inference → feature snapshot → scorer → write-back), a **unified event layer across 3 categories** for cross-agent escalation, **114 tests**, and CI at **80% branch coverage**

Open Health Intelligence (ClarityAtlas)

Sep 2025 – May 2026

Co-Founder & Technical Lead

Baltimore, MD

- Architected and shipped a **full-stack iOS application end to end** (Swift, SwiftUI; Supabase/PostgreSQL backend), piloted with **300+ Johns Hopkins students**; earned a **non-binding \$75K investment indication** and **presented to JHU President Ron Daniels and DSAI leadership**
- Designed a **tappigraphy-based sleep-estimation model** inferring sleep windows from phone-interaction and motion signals at **~90% accuracy**, replacing reliance on self-report or wearable hardware
- Owned the **entire backend solo**: PostgreSQL schema for longitudinal per-user data, **Supabase Edge serverless functions**, a **weighted Euclidean similarity-matching service**, and **8 geofenced triggers** driving context-aware push notifications

SnapSort

Jan 2024 – Aug 2025

Founder & Sole Engineer

Phoenix, AZ

- Shipped a web application solo from zero to **300+ users** (Flask, AWS) on a **multi-service pipeline** orchestrating third-party APIs behind one request path; trained a **CNN in PyTorch** at **~95% PCK**, raised **\$5K**, led a team of 4

PROJECTS

Dispatch — Workflow Automation Agent | TypeScript, Playwright, SQLite, Express, React

2026

- Built a deterministic agent automating a multi-step workflow across **3 third-party platforms**, driven by a **SQLite-backed state machine** governing **8 pipeline states** as the single source of truth for every transition, with a **Playwright** execution layer and an **Express + React** operator console
- Gated every mutating capability behind **fail-closed feature flags** and enforced a **four-stage validation ladder** requiring proof against live systems, not fixtures; free-text and sensitive fields route to human review by construction

tSearch — Talent Discovery Engine | TypeScript, Node, Neo4j, OpenAI

2026

- Shipped a working **MVP in 4 days** for a problem scoped by the founder of Z Fellows, iterating through **2 rounds of feedback** into an introduction to his team
- Built **2,000+ candidate profiles** into a navigable **agent graph of 20+ nodes and edges** (Neo4j) traversable by collaboration relationship, with a **multi-stage pipeline** (identity resolution → graph expansion → scoring → digest) and rate-limit-handled GitHub API ingestion

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, C/C++, SQL, Swift, R

Backend & Data: FastAPI, Flask, Node, Express, REST APIs, PostgreSQL, SQLite, Neo4j, Pinecone, schema design, streaming pipelines, state machines, concurrency; React, Vite, SwiftUI

ML & Tools: PyTorch, scikit-learn, XGBoost, sequence models, learning-to-rank, RAG, vector search; Git/GitHub (PRs, code review), CI, automated testing, observability, Playwright, AWS